

QUICKSHOW

Complete Website Flow, Web Pages, Endpoints & Functionalities

Project Blueprint

This document is the working blueprint for the QuickShow website: customer flow, admin flow, web pages, backend endpoints, functionality, database objects and implementation order.

Layer	Approach
Frontend	Thymeleaf + HTML/CSS/JS
Backend	Spring Boot
Controller	Endpoint-focused controllers using ResponseEntity
Service	Business logic and repository coordination
Repository	JdbcTemplate + raw SQL only
Database	MySQL
Authentication	Customer/Admin login; Spring Security planned

Development strategy: define endpoint → controller → required service logic → existing repository → endpoint test → Thymeleaf page.

1. Overall Architecture

Browser / Thymeleaf page
↓
Controller
↓
Service
↓
Repository (JdbcTemplate + raw SQL)
↓
MySQL

Controllers receive requests and return responses. Services contain business logic and coordinate repositories. Repositories contain explicit SQL and database interaction.

HTTP response convention

Situation	Response
Successful GET	200 OK
Successful POST	201 CREATED
Successful PUT	200 OK
Successful DELETE	204 NO CONTENT
Missing resource	404 NOT FOUND
Invalid request	400 BAD REQUEST where implemented
Business/database conflict	409 CONFLICT where required

2. Customer Journey

HOME
↓
Browse Movies
↓
Movie Details
↓
Select Show
↓
Select Seats
↓
Booking Summary
↓
Payment
↓
Booking Confirmation
↓
My Bookings

The customer browses movies, chooses a show and seats, reviews the booking, completes payment and receives confirmation. Bookings can later be viewed from My Bookings.

3. Admin Journey

ADMIN LOGIN
↓
ADMIN DASHBOARD
■■■ Manage Movies
■■■ Manage Movie Cast
■■■ Manage Theatres
■■■ Manage Screens
■■■ Manage Seats
■■■ Manage Shows
■■■ Manage Bookings
■■■ Manage Customers

4. Public & Customer Web Pages

Page	URL	Functionality	Data	Backend
Home / Landing	/	Entry point; movie search; now showing; upcoming movies; movie cards; ratings; login/register.	Movie, Show	GET /api/movies; show endpoints planned
Movie Listing	/movies	Browse catalogue; filter/search by language, genre and rating.	Movie	GET /api/movies; language and genre/rating searches
Movie Details	/movies/{movieId}	Poster, title, director, language, genre, duration, release date, rating, certificate, description, cast and shows.	Movie, MovieCast, Show, Screen, Theatre	GET /api/movies/{id}; cast/show endpoints planned
Theatre Listing	/theatres	Browse theatres and location/contact information.	Theatre	GET /api/theatres (planned)
Theatre Details	/theatres/{theatreId}	Address, contact, screens, screen type, capacity and shows.	Theatre, Screen, Show	GET /api/theatres/{id}; screen endpoints planned
Login	/login	Customer/admin authentication and routing.	Customer/Admin	Security/authentication planned
Registration	/register	Create customer account and associated email/profile data.	Customer, CustomerEmail	POST /api/customers/register (planned)
Customer Profile	/profile	View/edit profile and access booking/password actions.	Customer, CustomerEmail	GET/PUT customer endpoints planned
Show Selection	/movies/{movieId}/shows	Choose date, theatre, screen, time and ticket price.	Movie, Theatre, Screen, Show	GET /api/movies/{movieId}/shows; GET /api/shows/{id} planned
Seat Selection	/shows/{showId}/seats	Seat layout, available/booked status, selected seats and total.	Show, Screen, Seat, ShowSeatAllocates	GET /api/shows/{showId}/seats planned
Booking Summary	/shows/{showId}/booking	Review movie, theatre, show, seats, price and total.	Customer, Show, Booking, BookedSeats	POST /api/bookings planned
Payment	/payment	Display amount/payment method; process or record payment.	Payment, Booking	POST /api/payments planned
Booking Confirmation	/booking/{bookingId}/confirmation	Booking ID, movie, theatre, screen, date, time, seats, amount and payment status.	Booking, BookedSeats, Payment	GET /api/bookings/{id} planned
My Bookings	/bookings	Upcoming/past bookings, status, seats and payment information.	Booking, BookedSeats, Payment, Show, Movie	GET /api/customers/{id}/bookings planned
Booking Details	/bookings/{bookingId}	Complete booking information and cancellation where allowed.	Booking, BookedSeats, Payment	GET /api/bookings/{id}; cancel endpoint planned

5. Admin Web Pages

Page	URL	Functionality	Data	Backend
Admin Login	/admin/login	Authenticate administrators.	Admin	Security/authentication planned
Admin Dashboard	/admin/dashboard	Overview counts, statistics and links to management modules.	All major modules	Dashboard endpoints planned
Movie Management	/admin/movies	List, add, edit and delete movies.	Movie	GET/POST/PUT/DELETE /api/movies...
Movie Cast	/admin/movies/{movieId}/cast	View/add/remove actors.	MovieCast	GET/POST/DELETE cast endpoints planned
Theatre Management	/admin/theatres	List, add, edit and delete theatres.	Theatre	GET/POST/PUT/DELETE /api/theatres...
Screen Management	/admin/theatres/{theatreId}/screens	Manage screens, screen type and capacity.	Screen, Theatre	GET/POST/PUT/DELETE /api/screens...
Seat Management	/admin/screens/{screenId}/seats	View/configure seats and layouts.	Seat, Screen	GET/POST/DELETE seat endpoints planned
Show Management	/admin/shows	Create/edit/delete shows; movie, screen, date, time, price and status.	Show, Movie, Screen	GET/POST/PUT/DELETE /api/shows...
Booking Management	/admin/bookings	View bookings, customers, shows, seats, amounts and status.	Booking, BookedSeats, Payment	GET/booking actions planned
Customer Management	/admin/customers	View customers and relevant booking information.	Customer, CustomerEmail, Booking	GET /api/customers... planned

6. Current Movie Endpoints

Method	Endpoint	Purpose
GET	/api/movies	Get all movies
GET	/api/movies/{id}	Get movie by ID
GET	/api/movies/search/language/{language}	Find movies by language
GET	/api/movies/search/genre/{genre}?minRating={rating}	Find by genre and minimum rating
POST	/api/movies	Create movie
PUT	/api/movies/{id}	Update movie
DELETE	/api/movies/{id}	Delete movie

7. Planned Endpoint Map

Feature	Main endpoints
Movie Cast	GET/POST/DELETE /api/movies/{movieId}/cast...
Theatre	GET/POST/PUT/DELETE /api/theatres...
Screen	GET/POST/PUT/DELETE /api/screens...
Seat	GET/POST/DELETE /api/screens/{screenId}/seats...
Show	GET /api/shows/{id}; GET /api/movies/{movieId}/shows; POST/PUT/DELETE /api/shows...
Show Seats	GET /api/shows/{showId}/seats

Feature	Main endpoints
Customer	POST /api/customers/register; GET/PUT /api/customers/{id}
Bookings	POST /api/bookings; GET /api/bookings/{id}; POST /api/bookings/{id}/cancel
Customer Bookings	GET /api/customers/{customerId}/bookings
Payment	POST /api/payments; GET /api/payments/{id}
Admin	Protected management endpoints over the resources above

8. Database Object → Website Feature

Object	Where it appears
Movie	Home, movie list/details, show selection, booking, admin movie management
MovieCast	Movie details and admin cast management
Theatre	Theatre list/details, show selection, admin theatre management
Screen	Theatre details, show selection, seat selection, admin screen management
Seat	Seat selection and admin seat management
Show	Movie details, show selection, seat selection, booking and admin show management
ShowSeatAllocates	Show-specific seat availability and booking logic
Customer	Registration, profile, booking and admin customer management
CustomerEmail	Registration/profile/email functionality
Booking	Booking summary, confirmation, My Bookings and admin booking management
BookedSeats	Booking summary, confirmation and My Bookings
Payment	Payment page, confirmation, booking details and admin booking management
Admin	Admin login and administration

9. Booking Transaction Flow

```
Customer selects show and seats
↓
Verify customer
↓
Verify show
↓
Verify requested seats
↓
Check ShowSeatAllocates status
↓
Create Booking
↓
Create BookedSeats
↓
Change allocations to BOOKED
↓
Process / record Payment
↓
COMMIT
```

Failure at a required step → ROLLBACK

The booking operation should eventually be one transaction so the system cannot leave a partial booking if a later operation fails.

10. Endpoint-First Development Order

Step	Feature	Backend flow
1	Movies	MovieController → MovieService → MovieRepository
2	Theatres	TheatreController → TheatreService → TheatreRepository
3	Screens	ScreenController → ScreenService → ScreenRepository
4	Seats	SeatController → SeatService → SeatRepository
5	Movie Cast	MovieCastController → MovieCastService → MovieCastRepository

Step	Feature	Backend flow
6	Shows	ShowController → ShowService → ShowRepository
7	Show Seat Allocation	Controller/service → allocation repository
8	Customer/Auth	CustomerController → CustomerService
9	Booking	BookingController → BookingService
10	Payment	PaymentController → PaymentService
11	Security	Spring Security and protected routes
12	Thymeleaf	Connect completed endpoints to UI
13	End-to-end testing	Browser/Postman + database verification